

DiscoverAI

Semantic Product Search for Health & Personal Care

A Review-Aware Search System on Amazon Reviews 2023, with Quality-Driven Re-ranking,
BART Summarization, and Domain Entity Extraction

Team: Mean-Squared-Terrors

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Final Project — Data Science in Action
LUISS Guido Carli — A.Y. 2025/2026
May 2026

1 Introduction

Online retail platforms carry catalogues of tens of thousands of products. When a shopper types a natural-language query, keyword-based engines perform a purely lexical match and rank by text similarity or popularity, remaining oblivious to the *intent* behind the words. Three failure modes are acute in Health & Personal Care: (i) *semantic intent* — “something to help me sleep at night” returns nothing useful when “sleep” is absent from product descriptions; (ii) *implicit exclusions* — “face cream without parabens” can return paraben-containing creams; (iii) *trust signals* — millions of customer reviews capture real-world product performance, yet lexical search ignores them.

DiscoverAI addresses these failures by building a *review-aware* semantic search system on the Amazon Reviews 2023 dataset (Health & Personal Care category). The system understands natural-language queries through dense embeddings, integrates the voice of customers alongside product text, and re-ranks results by quality signals.

Dataset. The system is built on the Amazon Reviews 2023 corpus restricted to Health & Personal Care. After cleaning, the catalogue consists of **7,647 products** and **318,258 cleaned reviews**; a balanced subset of **57,097 reviews** is used for embedding. Of those products, **7,593** are loaded into the searchable index (54 products are dropped during NB03 because every review fell below the 25-token threshold).

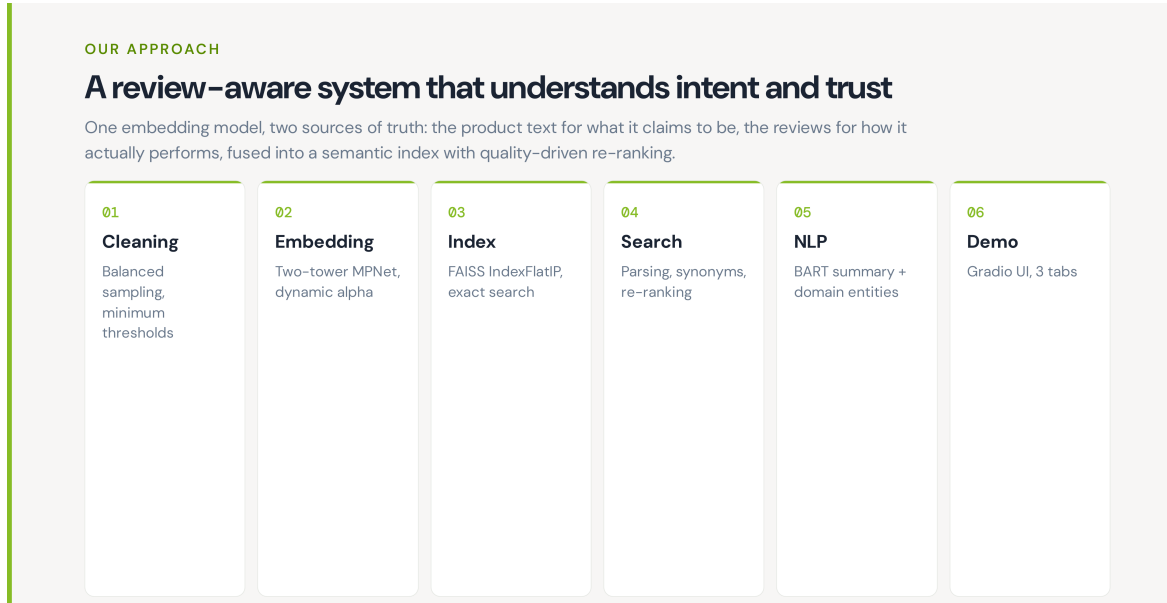


Figure 1: End-to-end *DiscoverAI* pipeline: six stages from raw data to the Gradio demo. Stage 01 (*Cleaning*) applies balanced sampling and minimum thresholds; Stage 02 (*Embedding*) runs two-tower MPNet with dynamic α ; Stage 03 (*Index*) loads all vectors into FAISS IndexFlatIP; Stage 04 (*Search*) handles parsing, synonym expansion, and re-ranking; Stage 05 (*NLP*) runs BART summarization and domain entity extraction; Stage 06 (*Demo*) exposes the full pipeline via a Gradio UI.

Approach and results. We propose a two-tower embedding architecture with dynamic alpha blending, an exact FAISS index, and an iterative search pipeline culminating in a hybrid BM25 + semantic retrieval system with Reciprocal Rank Fusion (RRF). Evaluated on a 37-query benchmark with pseudo-relevance judgements: **hybrid_v4** matches the strong BM25 baseline on overall NDCG@5 (0.535 vs 0.535) while *leading* on the open-ended *semantic intent* slice (0.581 vs 0.531). Pure semantic retrieval achieves the highest MRR (0.855). A three-layer guardrail rejects off-topic queries with **F1 = 0.957**.

2 Methods

2.1 Data Ingestion and Cleaning

Raw data. `Health_and_Personal_Care.json` and its metadata companion are loaded from the Amazon Reviews 2023 release. Each product carries title, brand, features, description, and a `details` dictionary; each review carries text, rating and `helpful_vote`.

Product-level filters. Products with fewer than **10 reviews** are discarded; reviews shorter than **25 tokens** are dropped.

Balanced rating sampling. 61.7% of raw reviews carry a five-star rating, which would flood the embedding with undifferentiated praise. Per-product caps $\{5:5, 4:3, 3:2, 2:3, 1:3\}$ are applied (top- k by `helpful_vote` inside each rating bucket). The five-star share drops to **46.2%**.



Figure 2: Rating distribution before (raw) and after (balanced) sampling. The rightmost chart highlights the reduction of the five-star share from 61.7% to 46.2%.

Product text construction. Title, brand, features, description and selected `details` keys (item form, skin type, active ingredients, ...) are concatenated, token-capped so the final text fits within MPNet’s 512-token limit.

2.2 Exploratory Data Analysis

EDA characterised rating distributions, review-length histograms, price buckets and helpful-vote patterns. Key findings: the 10-review minimum emerges from the long tail; reviews below 25 tokens are predominantly noise; price buckets (*budget*, *mid*, *premium*, *luxury*) are defined from price-quartile boundaries; only a handful of `details` keys carry consistent semantic value.

2.3 Embedding Architecture

Two-tower design. Each product gets two independent embeddings with `all-mpnet-base-v2` (768 d): a *product embedding* from the cleaned product text and a *review embedding* as a weighted average over the balanced reviews. The base weight of review r is

$$w_r^{\text{base}} = 0.70 \cdot \tilde{h}(r) + 0.30 \cdot \frac{|r_{\text{rating}} - 3|}{2}$$

where $\tilde{h}(r)$ is the log-normalised helpful-vote. Each base weight is then multiplied by a per-product *bias correction* $\text{clip}(\text{global_pct}/\text{local_pct}, 0.5, 3.0)$ that boosts rare rating bands within each product so that minority voices are not drowned out.

Dynamic alpha blending. $\mathbf{c} = \alpha \mathbf{p} + (1 - \alpha) \mathbf{r}$, normalised. Alpha is computed per product (Algorithm 8); on the indexed 7,593 products it has mean 0.560 and empirical range $[0.350, 0.732]$.

FAISS IndexFlatIP. Exact inner-product search, equivalent to cosine similarity after normalisation. At 7,593 products exact search adds negligible latency while guaranteeing zero missed neighbours; approximate alternatives (IVF, HNSW) become relevant only past tens of millions of items.

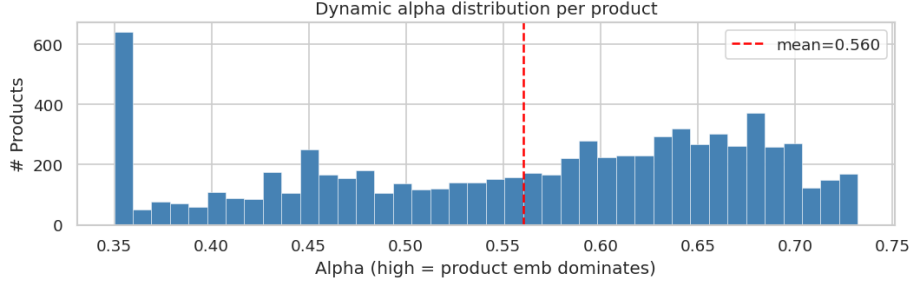


Figure 3: Empirical distribution of dynamic α across all 7,593 indexed products (mean = 0.560). Products with richer review evidence receive lower α , giving more weight to the customer voice.

2.4 Search and Re-ranking Pipeline

NB04 (`search.py`, `search_extended.py`) implements the retrieval pipeline. `search` retrieves **top-50** FAISS candidates, `search_v3` retrieves **top-80**, and `search_v4` retrieves **top-100** from both BM25 and FAISS.

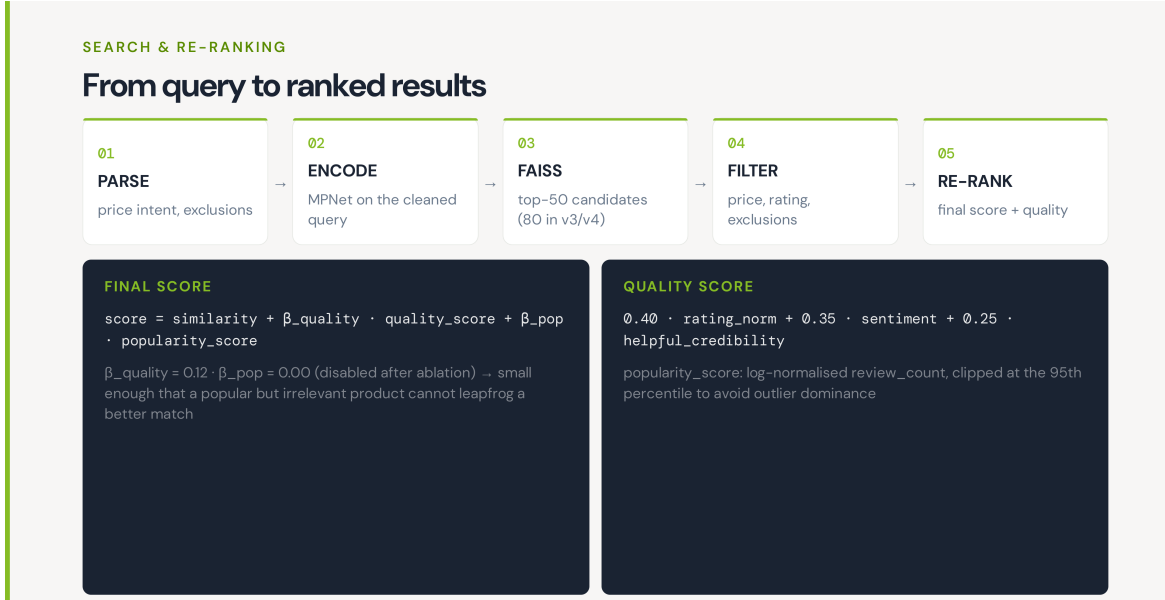


Figure 4: Query-to-results pipeline in `hybrid_v4`. Five stages: (1) *Parse* extracts price intent, exclusion terms, synonyms; (2) *Encode* runs MPNet on the cleaned query; (3) *FAISS* retrieves top-50/80/100 candidates depending on the search version; (4) *Filter* applies price, rating, and negation constraints; (5) *Re-rank* computes the final score using similarity, quality, and (disabled) popularity signals.

Re-ranking formula.

$$\text{score} = \text{sim} + \beta_q \cdot \text{quality} + \beta_p \cdot \text{popularity}, \quad \beta_q = 0.12, \beta_p = 0.00.$$

$$\text{quality} = 0.40 \cdot \bar{r}_{\text{norm}} + 0.35 \cdot \text{sentiment} + 0.25 \cdot \text{helpful_credibility}$$

β_q is deliberately small: a high-quality but irrelevant product cannot overtake a genuinely relevant one. $\beta_p = 0$ was the outcome of the ablation (Section 3.3).

Iterative versioning. `search` (top-50 + fixed re-rank) handles catalogue-style queries but fails on negation. `search_v2` adds negation patterns (“without X”, “no X”, “X-free”). `search_v3` adds synonym expansion, price-intent parsing, dosage normalisation and *adaptive* β_q : when the mean similarity of the 80 retrieved candidates falls below 0.65, β_q raises from 0.12 to 0.20. `hybrid_v4` combines BM25 and semantic retrieval through RRF (Algorithm 9); this is the system powering the Gradio demo.

2.5 Safety Layer

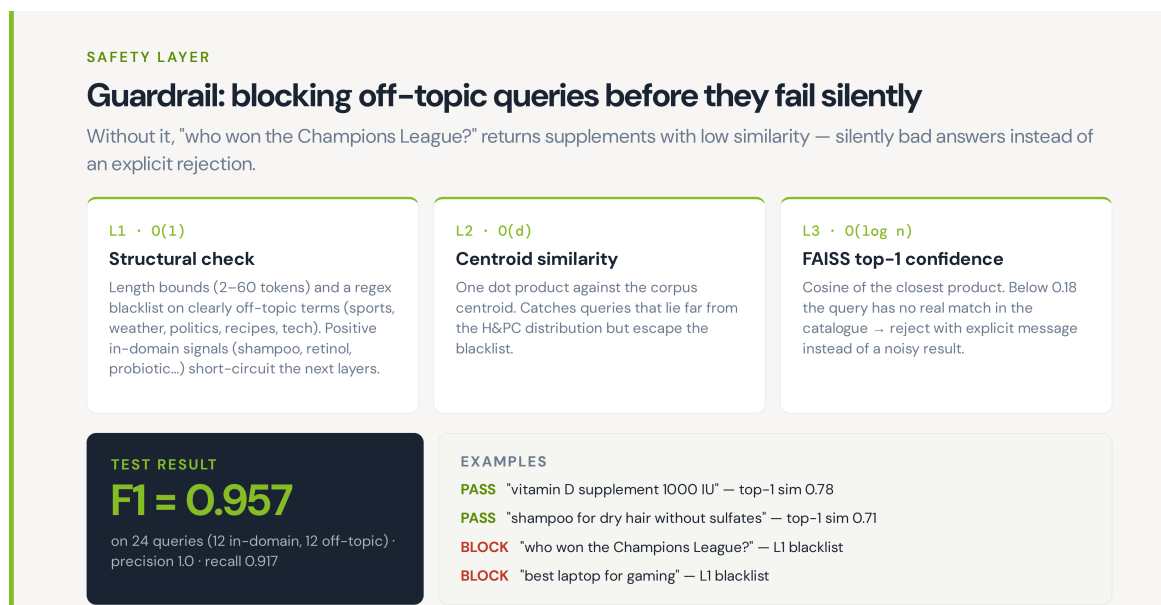


Figure 5: Three-layer guardrail for off-topic query rejection. L1 (*structural check*, $O(1)$) applies token-length bounds (2–60 tokens) and a regex blacklist; L2 (*centroid similarity*, $O(d)$) measures distance from the Health & Personal Care corpus centroid; L3 (*FAISS top-1 confidence*, $O(\log n)$) rejects queries whose closest product falls below cosine similarity 0.18. The system achieves $F1 = 0.957$ on a 24-query test set (precision = 1.000, recall = 0.917).

`src/guardrail.py` implements a three-layer cascade: (L1) structural check $O(1)$ — token-length bounds [2, 60] and a regex blacklist on off-topic domains; (L2) centroid similarity $O(d)$ — one dot product against the corpus centroid; (L3) FAISS top-1 confidence $O(\log n)$ — the closest product must have cosine similarity ≥ 0.18 . Positive in-domain signals (e.g. *shampoo*, *retinol*, *probiotic*) short-circuit the remaining layers.

2.6 Summarization and Entity Recognition

Reviews are split into positive ($\geq 4\star$) and negative ($\leq 2\star$) buckets. Informative sentences (≥ 12 words, not starting with “I”, < 300 chars) are deduplicated and fed to **BART-Large-CNN**, which produces an abstractive summary plus explicit *pros* and *cons* fields. An extractive fallback is implemented for products with very thin review evidence; on the current catalogue the BART model produced summaries for 7,645 of 7,647 products (99.97%; 2 products empty).

Entity extraction. Regex patterns cover active ingredients (retinol, niacinamide, hyaluronic acid, SPF, omega-3, ...), certifications (organic, vegan, cruelty-free, paraben-free, ...) and use-case targets. spaCy `en_core_web_sm` NER extracts brand mentions. `product_profiles.csv` merges the embedding index, summaries and entities into a single table consumed by the demo.

2.7 Evaluation Protocol

NB08 provides a formal benchmark: 38 hand-crafted queries with intent tags (*semantic*, *price*, *negation*, *dosage*, *brand*, *off-topic*). Pseudo-relevance judgements are assigned at four levels (0–3) via keyword matching on `product_text_base`. The single off-topic query (Q38) is excluded from the retrieval benchmark, leaving **37 scorable queries**. Metrics: NDCG@5, MRR, Recall@5, Precision@5, MAP@10.

Caveat. Pseudo-relevance via keyword matching mildly favours BM25, since it relies on the same tokens. A human-annotated relevance set would give a less biased estimate of semantic quality.

3 Results and Discussion

3.1 Embedding Space Geometry

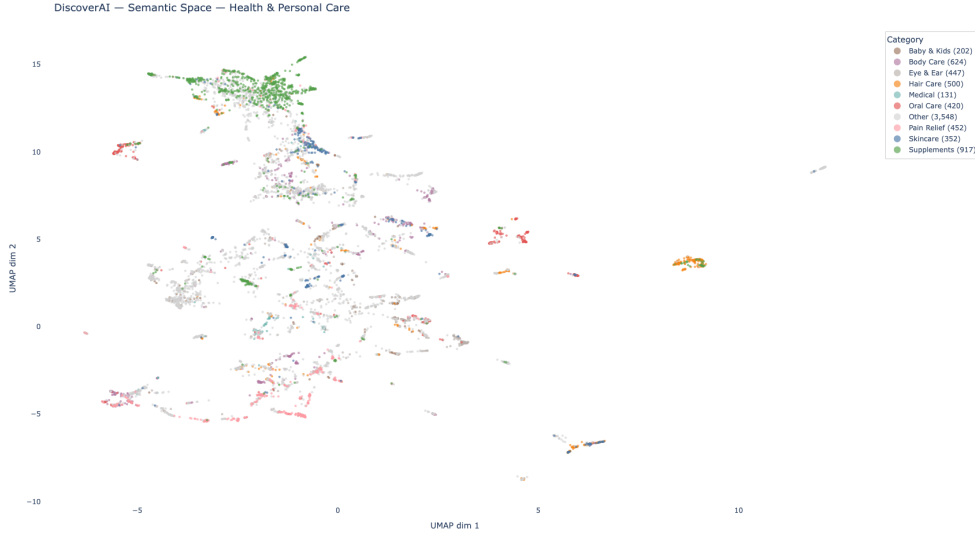


Figure 6: UMAP projection of the 7,593 combined product embeddings, coloured by inferred macro-category. Spatially coherent clusters emerge without any category supervision: *Supplements* form a tight group; *Hair Care* separates cleanly from *Skin Care*; *Pain Relief* sits at the intersection of supplements and topical products.

The 2-D UMAP projection of the 7,593 combined embeddings shows macro-categories emerging as spatially coherent clusters without any category supervision: *Supplements* form a tight group; *Hair Care* separates cleanly from *Skin Care*; *Pain Relief* sits at the intersection of supplements and topical products.

3.2 Retrieval Benchmark

Table 1 reports the five systems on 37 scorable queries (source: `src/io/eval_output/summary.csv`).

Table 1: Retrieval benchmark. Best value per metric in **bold**. 37 queries, $k = 5$.

System	NDCG@5	MRR	Recall@5	P@5	MAP
BM25 (baseline)	0.535	0.811	0.153	0.632	0.522
Semantic (no re-rank)	0.526	0.855	0.138	0.573	0.444
search (base)	0.517	0.780	0.135	0.568	0.425
search_v3	0.521	0.752	0.122	0.541	0.400
hybrid_v4 (demo)	0.535	0.784	0.119	0.557	0.438

`hybrid_v4` is statistically tied with BM25 on overall NDCG@5 (0.5348 vs 0.5353; absolute gap 5×10^{-4}); semantic retrieval without re-ranking is the strongest on MRR (0.855), meaning it most often surfaces the single most relevant product at rank 1. The contribution of `hybrid_v4` becomes clear in the per-intent breakdown (Table 2): it dominates the open-ended *semantic* slice while BM25 dominates the lexical-specific intents.

3.3 Ablation Study

Two ablations informed the final configuration:

Table 2: NDCG@5 by query intent (source: `per_intent_ndcg.csv`). Winner per row in **bold**.

Intent	BM25	hybrid_v4	search	search_v3	semantic
Brand	0.647	0.587	0.592	0.592	0.605
Dosage	0.906	0.762	0.597	0.663	0.550
Price	0.438	0.193	0.210	0.210	0.433
Negation	0.340	0.435	0.469	0.498	0.455
Semantic	0.531	0.581	0.562	0.553	0.544

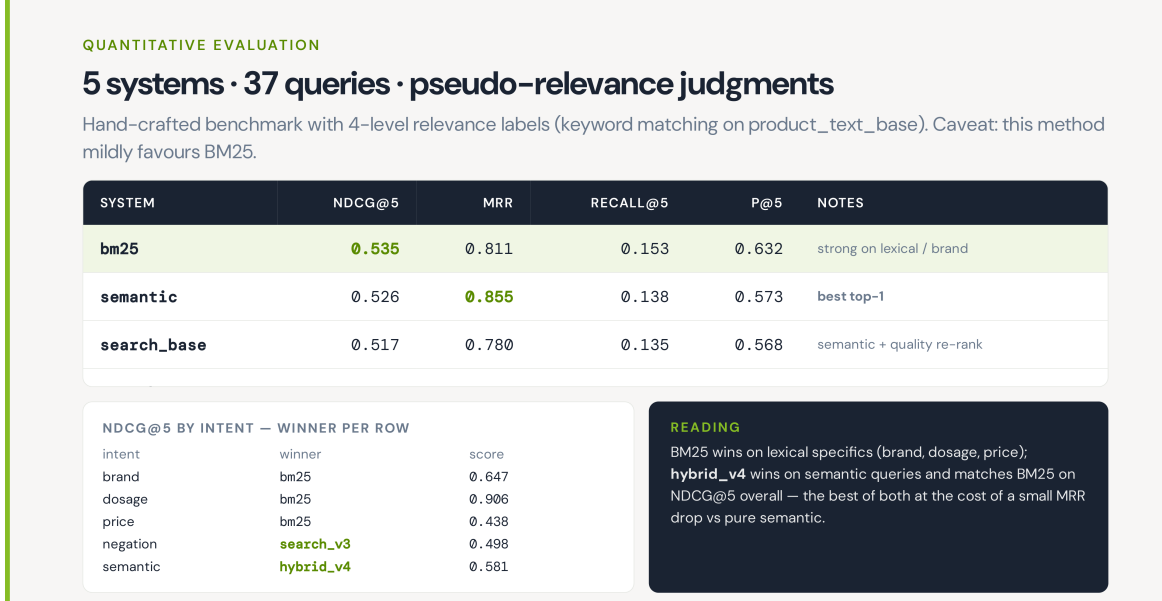


Figure 7: Retrieval benchmark summary and NDCG@5 per query intent. BM25 wins on lexical-specific intents (brand, dosage, price); **hybrid_v4** leads on open-ended semantic queries and matches BM25 on NDCG@5 overall, confirming that combining lexical and dense retrieval is the right production choice.

- **Popularity signal (β_p).** Enabling `popularity_score` (log-normalised review count, clipped at the 95th percentile) degraded MRR by $\sim 2.7\%$ at $\beta_p = 0.05$ ($0.7523 \rightarrow 0.7320$) and by $\sim 6.9\%$ at $\beta_p = 0.10$. The signal was disabled ($\beta_p = 0$) in all reported systems.
- **Adaptive β_q .** Raising β_q from 0.12 to 0.20 when the mean candidate similarity falls below 0.65 improved performance on ambiguous queries by shifting weight toward crowd-validated quality.

3.4 Safety Guardrail Evaluation

Tested on 24 queries (12 in-domain, 12 off-topic): precision = 1.000, recall = 0.917, **F1 = 0.957**. The single false negative was “*how to write a python decorator*”, a programming query that escaped the structural blacklist (no positive in-domain signal, but also no negative pattern matched). All 12 in-domain queries were correctly admitted.

3.5 Summarization and Entity Coverage

BART-Large-CNN produces a summary for 99.97% of products (7,645/7,647). Pros are extracted for 97.1% of summarised products, cons for 75.1%. Entity coverage by class: brand entities (via spaCy NER) cover 68.5% of the catalogue; certification labels reach 19.6%; active-ingredient

coverage is 8.8%, bounded by the specificity of the current regex catalogue. These structured facets power filter-based browsing within the demo’s search results.

3.6 Business-Oriented Results

Honest ranking — balanced sampling corrects the 5-star bias. **Constraint satisfaction** — negation filtering and synonym expansion preserve user intent. **Reduced decision friction** — BART summary cards (pros/cons) cut time-to-decision. **Brand safety** — the three-layer guardrail returns an explicit rejection instead of silently low-confidence results. **Scalability pathway** — dynamic alpha and RRF are catalogue-size-agnostic; only the FAISS index would need migration (`IndexFlatIP` $\rightarrow$ `IndexHNSW` / `IndexIVF`) past tens of millions of products.

4 Conclusions

Take-aways. *DiscoverAI* demonstrates that a two-tower architecture with dynamic alpha blending and hybrid BM25 + semantic retrieval via RRF is a robust foundation for review-aware product search. Three design choices proved most impactful: (i) keeping product text and review aggregations as separate towers rather than concatenating them; (ii) calibrating the quality re-rank weight small enough ($\beta_q=0.12$) that relevance always dominates popularity; (iii) treating the guardrail as a first-class system component rather than an afterthought. The NLP layer (BART + regex/spaCy NER) adds interpretable richness to each result without burdening retrieval latency.

Limitations. Pseudo-relevance judgements based on keyword matching introduce a mild evaluation bias in favour of BM25. The benchmark (37 queries) is small, and no A/B test with real users was conducted. BART inference latency ($\sim 1\text{--}2$ s per product batch) is not suitable for on-the-fly online summarization at scale and must be cached. The evaluation is limited to the Health & Personal Care vertical; portability to other categories remains untested.

Future work. (1) cross-encoder re-ranking for a two-stage retrieval pipeline; (2) fine-tuning `all-mpnet-base-v2` on click/purchase data; (3) pre-computing and caching all BART summaries at index-build time; (4) replacing pseudo-relevance with human relevance judgements; (5) multi-category expansion beyond Health & Personal Care.

References

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A Code Description

All source code lives under `src/mean_squared_terrors/`. Notebooks 01–08 orchestrate the pipeline in order; each notebook imports from the corresponding module in `src/`.

Table 3: Source modules and their roles.

Module	Role
<code>config.py</code>	All tunable parameters
<code>cleaning.py</code>	Text normalisation, balanced sampling
<code>eda.py</code>	Plotting helpers (NB02)
<code>embedding.py</code>	MPNet encoding, dynamic α , FAISS
<code>embedding_viz.py</code>	UMAP projection, Plotly scatter
<code>search.py</code>	v1–v3 search, negation/synonym/price
<code>search_extended.py</code>	hybrid_v4: BM25 + semantic + RRF
<code>summarization.py</code>	BART, spaCy NER, profile merge
<code>guardrail.py</code>	Three-layer off-topic detection
<code>vector_db.py</code>	ChromaDB persistence
<code>eval_set.py</code>	38 evaluation queries
<code>eval_metrics.py</code>	NDCG@K, MRR, Recall@K, P@K, MAP@K
<code>run_evaluation.py</code>	End-to-end evaluation runner

Algorithm 1 — Dynamic α for a single product

Input: n (number of balanced reviews), $h_{\max}$ (max helpful vote).

1. $n_{\text{norm}} \leftarrow (\min(n, 16) - 3)/13$
2. $h_{\text{norm}} \leftarrow \min(h_{\max}, 20)/20$
3. $q \leftarrow 0.6 \cdot n_{\text{norm}} + 0.4 \cdot h_{\text{norm}}$
4. $\alpha \leftarrow \alpha_{\max} - q \cdot (\alpha_{\max} - \alpha_{\min})$ with $\alpha_{\min} = 0.35$, $\alpha_{\max} = 0.70$
5. **return** α

Figure 8: Dynamic α computation (label: `alg:alpha`).

Algorithm 2 — hybrid_v4: BM25 + Semantic with RRF

Input: query q ; constant $k = 60$; cutoff $N = 100$.

1. $R_{\text{bm25}} \leftarrow \text{BM25_topN}(q)$
2. $R_{\text{sem}} \leftarrow \text{FAISS_topN}(\text{MPNet}(q))$
3. **for all** $d \in R_{\text{bm25}} \cup R_{\text{sem}}$ **do**
 $\text{score}(d) \leftarrow \sum_{i \in \{\text{bm25}, \text{sem}\}} 1/(k + \text{rank}_i(d) + 1)$
4. **end for**
5. **return** top- k products sorted by score (after quality re-rank)

Figure 9: hybrid_v4 fusion via Reciprocal Rank Fusion (label: `alg:rrf`).

B Author Contribution and Generative AI Statement

B.1 — Author Contributions (CRediT taxonomy)

Member	E-mail	Role(s)
Filippo Nannucci (delegate)	<code>filippo.nannucci@studenti.unipi.it</code>	Data Curation (lead); Visualization; Project Administration; Writing — Review & Editing. Primary author of <code>02_EDA.ipynb</code> ; built the Gradio demo (NB06); managed final submission and notebook structure alignment.
Daniele Giovanardi	<code>daniele.giovanardi@studenti.unipi.it</code>	Conceptualization ; Methodology; Software; Formal Analysis. Designed and implemented the two-tower MP-Net + FAISS embedding with dynamic alpha, the search pipeline ($v2 \rightarrow v3 \rightarrow v4$ with hybrid RRF), ChromaDB integration, guardrail, UMAP visualisation, and all empirical tuning decisions ($\beta_p = 0$, <code>min_rating=None</code> , $\alpha \in [0.35, 0.70]$ design target).
Gabriele Loreti	<code>gabriele.loreti@studenti.unipi.it</code>	Software ; Resources; Project Administration; Writing — Original Draft. Owned reproducibility and infrastructure: pinned <code>requirements.txt</code> , <code>.gitignore</code> , package structure, <code>src/io/</code> scaffolding, and the README.

B.2 — Generative AI Statement

In compliance with `project_rulez.pdf`, we declare the following responsible use of Generative AI tools during development. All AI-assisted output was treated as a draft to be reviewed, executed, and validated by the team before inclusion.

Models employed.

1. **Claude Sonnet (Anthropic)** — coding assistant for the modules under `src/mean_squared_terrors/`: refactoring suggestions (e.g. vectorisation of `compute_review_weights` in `embedding.py`), drafting docstrings, expansion of regex catalogues in `summarization.py`, translation of code comments into English, and scaffolding of unit test cases.
2. **Claude Opus (Anthropic)** — higher-level design and methodological discussion: brainstorming the two-tower product/review scheme with dynamic alpha blending (NB03), comparing `search_v2 / v3 / v4`, discussing Reciprocal Rank Fusion, and reviewing the structure of the technical report.
3. **GitHub Copilot** — in-IDE autocomplete while writing modules in `src/`. Suggestions were inspected line by line and edited before commit; no whole function was committed verbatim.

Validation. All AI-generated code was executed end-to-end via notebooks 01–08 and validated against the outputs of each pipeline stage. Factual claims about models and metrics (`all-mpnet-base-v2`, BART-Large-CNN, BM25, NDCG, MRR, MAP, RRF) were cross-checked

against the original papers and the official library documentation before being written into the report.

Scope of human authorship. The 38 hand-crafted benchmark queries in `eval_set.py`, the pseudo-relevance judgments, and the empirical tuning decisions ($\beta_{\text{pop}} = 0.0$, `min_rating=None`, the $[0.35, 0.70]$ design target for dynamic α) are the result of the team’s own analysis and design choices, not LLM output.

No GenAI tool was used to fabricate data, simulate experimental results, or produce content that the team cannot personally explain.